LICENCE TO DIVERT WATER
ALBERTA ENERGY REGULATOR
WATER ACT, R.S.A. 2000, c.W-3, as amended

LICENCE NO.: 00458425-00-00

APPLICATION NO.: 001-00458425

FILE NO.: 00458425

PRIORITY NO.: 2019-11-29-002

EFFECTIVE DATE: May 26, 2020

EXPIRY DATE: May 25, 2030

SOURCE OF WATER: Blindman River

POINT OF DIVERSION: NE-03-040-01-W5M

POINT OF USE: As shown in Schedule 3

LICENSEE: VESTA ENERGY LTD.

Pursuant to the *Water Act*, R.S.A. 2000, c.W-3, as amended and the South Saskatchewan Basin Water Allocation Regulation, as amended/Crown reservation/Approved Water Management Plan, a licence is issued to the Licensee to:

> operate a works and to divert up to 1,000,000 cubic metres of water annually at a maximum rate of diversion of 0.5 cubic metres per second from the source of water for the purposes of drilling and hydraulic fracturing and Associated Operations

subject to the attached terms and conditions.

Rieanne Graham
Manager, Regulatory Applications Branch
Alberta Energy Regulator

May 26, 2020

DEFINITIONS

1.0 All definitions from the Act and the Regulations apply except where expressly defined in this licence.

1.1 In all parts of this licence:

(a) “Act” means the Water Act, RSA 2000, c. W-3, as amended;

(b) “Application” means the written submissions to the Director in respect of Application number 001-00458425 and any subsequent Applications for amendments of licence No. 00458425-00-00;

(c) “Associated Operations” means directly related ancillary surface operations/activities which include access roads and lease site construction and maintenance, dust suppression, and operation of infrastructure related to hydraulic fracturing;

(d) “Director” means an authorized employee of the Alberta Energy Regulator;

(e) “Point(s) of Diversion” means the location(s) where water is diverted from the source of water;

(f) “Point of Use” means the location(s) in which the diverted water is used by the Licensee for the licenced purpose;

(g) “Point of Use Area” means the area identified in Schedule 3 and includes the Sub-surface Mineral Lease Area(s) with a surface buffer and existing permanent water management infrastructure including but not limited to reservoirs and freshwater pipelines;

(h) “Regulations” means the Regulations, as amended, enacted under the authority of the Act;

(i) “Sub-surface Mineral Lease Area(s)” means the area of the mineral lease(s) issued under the Mines and Minerals Act that grants the right to remove a mineral as identified in Schedule 3;

(j) “Volume Estimate” means a technical evaluation including but not limited to pump capabilities, water metres, or volume per truck load multiplied by number of loads;

(k) “Water Use Reporting System” means the secure internet website provided by Alberta Environment at https://www.alberta.ca/water-use-reporting-system.aspx for submitting measuring and monitoring results electronically to the Director;

(l) “Water Conservation Objective” means the amount and quality of water necessary for the protection of a natural water body or its aquatic environment; and

(m) “Instream Objective” means regulated minimum flows that should remain in the river via dam operations or as a restriction on licences for the protection of the aquatic environment.

GENERAL

2.0 The Licensee shall immediately report to the Director by telephone any contravention of the terms and conditions of this licence at 1-780-422-4505.

2.1 The terms and conditions of this licence are severable. If any term or condition of this licence is held invalid, the application of such term or condition to other circumstances and the remainder of this licence shall not be affected thereby.

2.2 The Licensee shall not deposit or cause to be deposited any substance in, on or around the source of water that has or may have the potential to adversely affect the source of water.

2.3 Within six months after permanently ceasing operation of the works or diversion of the water, the Licensee shall submit to the Director:

(a) an Application for the decommissioning of the works; or

(b) a proposal and justification for continued use of the works.

2.4 The Licensee shall comply with the terms and conditions of the “Water Use Reporting System User Consent”.

2.5 The Director may amend this *Water Act* licence to:

(a) reduce the volume of the allocation;

(b) change the rate of the diversion;

(c) change the timing of the diversion; or

(d) shorten the expiry date of the licence

upon a minimum of 60 days written notice to the Licensee, if the licensed allocation of water is not being used as stated in the Application or in accordance with the authorized purpose in this licence.

2.6 Upon receiving notice from the Director pursuant to 2.5, the Licensee shall provide to the Director, within 30 days or the time period specified by the Director’s notice;

(a) information on the current and future water needs;

(b) an overview of the Licensee's development plan including any updates or changes that have occurred or are planned that were not included in the Application, and

(c) any other information requested by the Director.

2.7 The Director may amend this *Water Act* licence to:

(a) reduce the volume of the allocation;

(b) change the rate of the diversion;

(c) change the timing of diversion;

(d) modify Schedule 1: Diversion Table; or

(e) Instream Objectives and/or Water Conservation Objectives;

upon a minimum of 60 days written notice to the Licensee, if the diversion of water authorized by this *Water Act* licence results in an adverse effect to the aquatic environment or to other users and Licensees under the *Water Act*, that was not reasonably foreseeable at the time the licence was issued or the diversion restriction values in Schedule 1: Diversion Table, Instream Objectives and/or Water Conservation Objectives are updated by the Government of Alberta.

2.8 The Sub-surface Mineral Lease Area under this licence cannot be modified, unless approved by the Director through written authorization or licence amendment.

2.9 The Licensee shall provide advance written notice to the Director at least 90 calendar days prior to any proposed changes to the Schedule 3 map and permanent water management infrastructure referenced in Schedule 3.

DIVERSION OF WATER

3.0 This licence is appurtenant to the land at the points of diversion at:

(a) NE-03-040-01W5M, as shown in Schedule 2.

3.1 The Licensee shall divert water only for the purposes of drilling and hydraulic fracturing of oil and gas wells and Associated Operations.

3.2 The Licensee shall divert water only from the source of water specified in this licence.

3.3 The Licensee shall divert water only from the following Point(s) of Diversion:

(a) Blindman River at NE-03-040-01W5M, as shown in Schedule 2.

3.4 The Licensee shall divert water only to the Point of Use Area as shown in Schedule 3.

3.5 The works used to divert the water authorized by this licence shall include, at a minimum, all of the following:

(a) the temporary pumps and hoses/pipelines used to divert water from the sources of water, including lay flat hoses;

(b) the fish screen on the source of water pump intake that meets the criteria in the *Interim Code of Practice: End-of-pipe fish protection screens for small water intakes in freshwater* as amended from time to time (Department of Fisheries and Oceans Canada http://www.dfo-mpo.gc.ca/pnw-ppe/codes/screen-ecran-eng.html); and

(c) the permanent water management infrastructure listed in Schedule 3.

3.6 The Licensee shall cause any water entering a pump, located on source of water, to first pass through the screen required in 3.5 (b).

3.7 The Licensee shall inspect and maintain the fish screen to ensure it meets the 'Screen Maintenance and Cleaning' criteria in the *Interim Code of Practice: End-of-pipe fish protection screens for small water intakes in freshwater*' as amended from time to time (Department of Fisheries and Oceans Canada, http://www.dfo-mpo.gc.ca/pnw-ppe/codes/screen-ecran-eng.html).

3.8 The Licensee must follow the *"Decontamination Instructions for Industrial and Construction Operations"* in the *"Decontamination Protocol for Watercraft and Equipment",* as amended from time to time, for all equipment that are used within the bed of the source of water:

(a) prior to operations;

(b) during operations when moving equipment across a Hydrological Unit Code (HUC) Level 6 watershed boundary indicated on the "Decontamination Risk Map" or the "Hydrological Unit Code Level 6 (HUC 6) Shape file"; and

(c) upon completions of operations;

or as otherwise authorized in writing by the Director.

3.9 The Licensee must provide a completed Decontamination Record form to the Director upon request.

3.10 Prior to diverting any water from the source of water, the Licensee shall equip the point of diversion with a meter, which measures:

(a) cumulatively, the quantity of all water diverted during the reporting period; and

(b) the instantaneous rate of diversion.

3.11 The Licensee shall maintain each measuring device referred to in 3.10 when diverting any water.

3.12 The Licensee shall calibrate each measuring device referred to in 3.10 in accordance with manufacturer's specifications.

3.13 The Licensee shall not divert more than 1,000,000 cubic metres of water per calendar year for the purpose of this licence.

3.14 The Licensee shall not divert from the source of water at a rate of diversion greater than:

(a) 0 cubic meters per second, when the flow in the Blindman River is equal to or less than the Instream Objective of 0.16 cubic meters per second;

(b) 10% of the rate of flow in the Blindman River when the flow in the Blindman River is greater than 0.16 cubic meters per second but less than or equal to 0.5 cubic meters per second;

(c) 15% of the rate of flow in the Blindman River up to a maximum diversion rate of 0.5 cubic meters per second; when the flow in the Blindman River is greater than 0.5 cubic meters per second.

3.15 The Licensee shall divert the water authorized by this licence only when there is sufficient flow in:

(a) the Red Deer River to meet or exceed the Water Conservation Objective; and

(b) the Blindman River to meet or exceed the Instream Objective (IO)

as specified in 3.16 and 3.17 respectively.

3.16 The Water Conservation Objective for the Red Deer River is a rate of flow in the Red Deer River that is 45% of the natural rate of flow or 16 cubic meters per second, whichever is greater at any point in time.

3.17 The Instream Objective for the Blindman River is a rate of flow in the Blindman River of 0.16 cubic meters per second.

3.18 Unless authorized in writing by the Director, the Red Deer River Water Conservation Objective is to be met:

(a) at the Water Survey of Canada Station 05CC002 (Red Deer River at Red Deer); during the open water season; or

(b) at the Water Survey of Canada Station 05CB007 (Dickson Dam Tunnel Outlet) during the winter ice cover season;

(c) when diverting any water.

3.19 Unless authorized in writing by the Director, the Blindman River Instream Objective is to be met:

(a) at the Water Survey of Canada Station 05CC001 (Blindman River near Blackfalds) when it is operational; or

(b) at the point of diversion when the WSC station 05CC001 is not operational;

(c) when diverting any water.

3.20 The Licensee shall not divert water when the water temperature in the source of water exceeds 22°C.

MONITORING AND REPORTING

4.0 Unless otherwise authorized in writing by the Director, the Licensee shall:

(a) measure the total volume of water diverted each month, from the source of water, using the meter specified in 3.10(a);

(b) measure the rate of diversion on a continuous basis, from the source of water, using the meter specified in 3.10(b);

(c) measure or provide a Volume Estimate of the total volume of water used for each Associated Operation and the water use per well licenced under the *Oil and Gas Conservation Act*;

(d) record the surface legal land location(s) where diverted water is being used, and the purpose/operation for which it is being used;

(e) when applicable, measure the total volume of water released to the environment;

(f) for the calendar year, compare the water diversion and usage as per 4.0 (a) to the estimates submitted with the Application and any previous reporting;

(g) for the upcoming calendar year, prepare an overall estimate of the monthly water diversion and usage that is expected under this licence;

(h) record any changes to the Sub-surface Mineral Lease Area(s) and well density plan; and

(i) prepare an assessment of any changes to the original schedule of development and the timelines required to complete the project.

4.1 When the Water Survey of Canada Station No. 05CC001 (Blindman River near Blackfalds) is operational, the Licensee shall monitor the rate of flow of water in the source of water:

(a) at the WSC Station No. 05CC001;

(b) daily during diversion;

or as otherwise authorized in writing by the Director.

4.2 When the Water Survey of Canada Station No.05CC001 (Blindman River near Blackfalds) is not operational, the Licensee shall measure the rate of flow of water in the source:

(a) at the point of diversion;

(i) a minimum of weekly when the measured rate of flow is less than or equal to 0.5 cubic metres per second;

(ii) a minimum of bi-weekly when the measured rate of flow is greater than 0.5 cubic meters per second.

during diversion, or as otherwise authorized in writing by the Director.

4.3 The Licensee shall monitor the rate of flow in the Red Deer River a minimum of bi-weekly during diversion.

4.4 Between June 1 and October 1, the Licensee shall measure the temperature of the source of water the point of diversion, at a minimum of the following:

(a) prior to commencing diversion of water to establish a baseline temperature; and

(b) if the baseline temperature:

(i) is greater than 19°C, measure the temperature hourly during diversion; or

(ii) any previous temperature measurement is less than 19°C, measure the temperature daily during diversion.

4.5 The Licensee shall record and retain all of the following information for a minimum of 5 years after being collected:

(a) the place, date and time of all monitoring and measuring;

(b) the results obtained pursuant to 4.0, 4.1, 4.2, 4.3 and 6.1; and

(c) the name of the individual who conducted, or who was responsible for, the monitoring, measuring and sampling stipulated in (a) and (b) and 6.1.

4.6 The Licensee shall report to the Director the results of the measuring and monitoring required in 4.0 (a) using the “Water Use Reporting System” and any other information required in writing by the Director.

4.7 The Licensee shall submit the information in 4.5 on or before the end of the month following the month in which the information is based upon was collected.

4.8 The Licensee shall report to the Director the results of the measuring and monitoring required in 4.0 (c) to (i), 4.1, 4.2. 4.3 and 6.1 using the "Water Use Reporting System" and any other information required in writing by the Director.

4.9 The Licensee shall submit the information in 4.7 on or before March 31st of the year following the year in which the information was obtained.

COMPLAINT INVESTIGATION

5.0 The Licensee shall:

(a) investigate all written complaints accepted by the Director relating to allegations of surface water and groundwater interference as a result of the diversion of the water or operation of the works; and

(b) provide a written report to the Director, within a time specified in writing by the Director, detailing the results of the investigation relating to the complaint accepted by the Director in 5.0(a).

5.1 The Licensee shall satisfy the Director that the report submitted pursuant to 5.0(b) has identified remedial and/or mitigative measures relating to the alleged interference.

WATER RELEASE

6.0 The Licensee shall not release any diverted water from the works if any substances from any source have been added or co-mingled in the diverted water, unless authorized in writing by the Director.

6.1 The Licensee can release water, diverted under this licence, back to the environment if the following criteria are met:

(a) the Licensee has investigated the potential to use the excess water diverted under this licence in other oil and gas operations and documented the uses investigated and the reasons why these were not feasible;

(b) the return is made to a watershed where the original diversion took place;

(c) consent is obtained from the landowner or land manager at the point of release location;

(d) an assessment of the potential impacts to downstream stakeholders is conducted, and those stakeholders are notified, when appropriate;

(e) any excess fresh water that is discharged onto a land surface (frozen or unfrozen soil conditions) shall be in a controlled fashion that does not result in flooding or erosion, or uncontrolled direct flow into a water body or road ditch;

(f) all applicable erosion prevention and other environmentally precautionary measures are implemented;

(g) water must not be released to land that has the following characteristics:

(i.) soil with an electrical conductivity greater than 4 dS/m; or

(ii.) soil with a sodium adsorption ratio greater than 9.

(h) all applicable protocols and recommended practices to prevent the spread of aquatic invasive species are implemented;

(i) water must meet the release criteria as indicated in Schedule 4: Excess Water Release Criteria; and

(j) the provision of any other information as requested in writing by the Director.

6.2 If the excess water is not suitable for surface water discharge, alternative disposal options must be met according to existing procedures and guidelines.

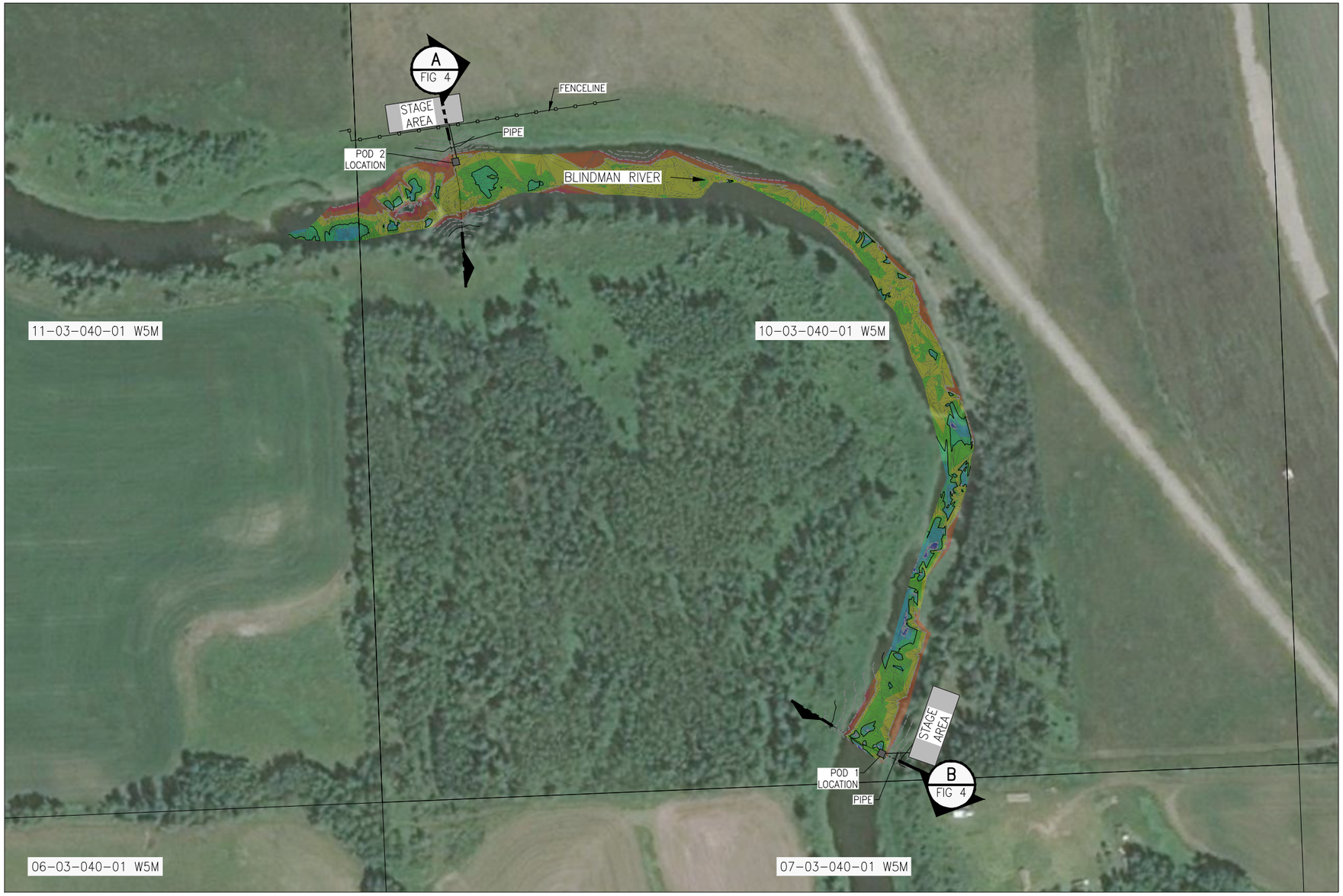
Rieanne Graham
Manager, Regulatory Applications Branch
Alberta Energy Regulator

May 26, 2020

SCHEDULE 1: Diversion Table

Criteria		**Minimum Monitoring/Measuring Frequency**	
Flow in Blindman River* (m^3/s)	**Maximum Rate of Diversion (m^3/s)**	**When using Station 05CC001**	**When measuring flows at POD**
0.16 ≥ Flow	0 (No Diversion allowed)	Daily	Weekly
0.16 < Flow ≤ 0.5	Up to 10% of the flow, provided the diversion does not decrease the Blindman River flow* to 0.16 m^3/s or lower		
Flow > 0.5	15% of the flow up to a maximum of 0.5 m^3/s		14 days

**Based on monitored flow at WSC Station 05CC001 (Blindman River at Blackfalds) when operational and measured flow at POD when WSC Station 05CC001 is not operational.*



LEGEND:

SURVEY GROUND (2.5m INTERVAL) (NOTE 1)

SURVEY GROUND (0.5m INTERVAL) (NOTE 1)

FENCELINE

Water Depths Table		
Minimum Depth	Maximum Depth	Color
0.00	0.25	
0.25	0.50	
0.50	0.75	
0.75	1.00	
1.00	1.25	
1.25	1.50	
1.50	1.75	
1.75	1.86	

Schedule 2 - POD Locations
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0 20 40 60 80 100m

SCALE 1 : 2,000

UTM NAD83 ZONE 11

NOTES:

1. BATHYMETRY SURVEY FROM INTEGRATED SUSTAINABILITY; DATED 2019-07-31.
2. SURVEY DATA NOT REFERENCED TO SPECIFIC MONUMENT.

PREPARED BY

INTEGRATED SUSTAINABILITY

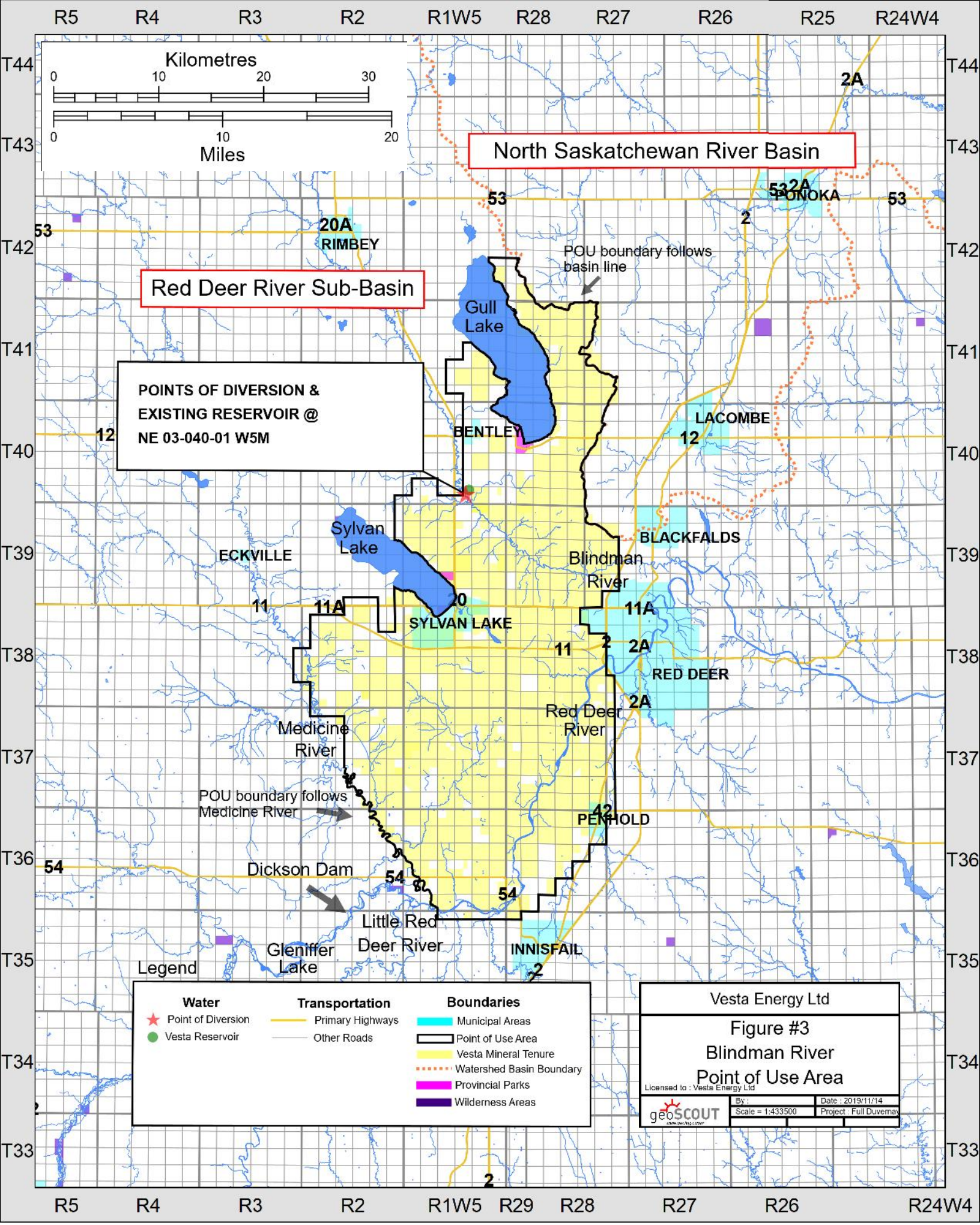
CLIENT

VESTA

VESTA ENERGY LTD.
TERM LICENSE APPLICATION - BLINDMAN RIVER
PLAN VIEW OF PROPOSED INTAKE SYSTEM SET-UP

DRAWN BY	CHECKED BY	APPROVED BY	DATE (YYYY-MM-DD)
H. ISIP	J. FRYERS	F. BHUIYAN	2020-05-25
SCALE	PROJECT NO.	FIGURE NO.	REVISION
1:2,000	CP19-VES-01-00	FIGURE 2	0

BY: CHERYL DORMAN, PLOTTED: 2020-05-25 11:19 AM, FILENAME: W:\Vesta Energy Ltd\CP19-VES-01-00\58 Civil-Earthworks\SK\CP19-VES-01-00-Plan and Profile.dwg


Kilometres
0 10 20 30
0 10 20
Miles
North Saskatchewan River Basin
Red Deer River Sub-Basin
POU boundary follows basin line
POINTS OF DIVERSION & EXISTING RESERVOIR @ NE 03-040-01 W5M
Gull Lake
Sylvan Lake
Blindman River
Red Deer River
Medicine River
POU boundary follows Medicine River
Dickson Dam
Little Red Deer River
Gleniffer Lake
RIMBEY
BENTLEY
LACOMBE
PONOKA
BLACKFALDS
ECKVILLE
SYLVAN LAKE
RED DEER
PENHOLD
INNISFAIL
Legend
Water
Point of Diversion
Vesta Reservoir
Transportation
Primary Highways
Other Roads
Boundaries
Municipal Areas
Point of Use Area
Vesta Mineral Tenure
Watershed Basin Boundary
Provincial Parks
Wilderness Areas
Vesta Energy Ltd
Figure #3
Blindman River
Point of Use Area
Licensed to : Vesta Energy Ltd
geoSCOUT
By :
Scale = 1:433500
Date : 2019/11/14
Project : Full Duvernay

SCHEDULE 4: Excess Water Release Criteria

1. Ensure a representative sample of the excess water diverted under this licence meets the following criteria prior to release, or as otherwise authorized in writing by the Director for cases where natural levels are above the release criteria:

   (a) pH between 6 and 9;

   (b) electrical conductivity (EC) ≤ 2 dS/m;

   (c) for land application cases only, sodium adsorption ratio (SAR) < 4;

   (d) for receiving water body cases only, total suspended solids < 50 mg/l;

   (e) chloride ≤ 120 mg/L;

   (f) sulphate ≤ 500 mg/L;

   (g) nitrate ≤ 3 mg/L; and

   (h) no visible hydrocarbon sheen.